PROBLEM STATEMENT 01

LEXGUARD

AI Rights & Contract Intelligence System

Background

In today's digital and professional ecosystem, individuals and organizations routinely accept legally binding agreements without fully understanding their implications. Employment contracts, vendor agreements, subscription terms, rental agreements, insurance policies, platform terms of service, and privacy policies often contain complex legal language that is difficult for non-specialists to interpret.

These agreements may include restrictive clauses, hidden liabilities, broad intellectual property transfers, automatic renewals, one-sided arbitration mechanisms, unfavorable termination conditions, or excessive data collection practices. Such clauses can significantly impact an individual's financial security, employment flexibility, privacy rights, and legal protections.

While several legal technology platforms provide basic document summarization or template generation, there remains a strong need for intelligent systems capable of identifying contractual risks, reasoning about their practical implications, and presenting insights in an understandable and transparent manner.

As digital agreements continue to increase in complexity and scale, there is growing demand for accessible AI-driven legal intelligence systems that improve awareness, transparency, and informed decision-making.

Problem Statement

Design and develop an AI-powered contract intelligence platform capable of analyzing legal and quasi-legal documents to identify potentially harmful, exploitative, ambiguous, or high-risk clauses before users agree to them.

The system should extract and classify important clauses, evaluate contractual risks, reason about possible real-world implications, and provide interpretable explanations from the perspective of the affected individual or organization.

The platform should support multiple categories of agreements and provide users with meaningful legal awareness rather than simple text summarization.

Objectives

The proposed solution should aim to:

- Analyze uploaded legal documents and extract meaningful contractual clauses
- Identify hidden liabilities, unfavorable obligations, and one-sided legal conditions
- Detect ambiguous or potentially exploitative language
- Highlight privacy, financial, employment, intellectual property, and compliance-related risks
- Provide understandable explanations of contractual implications in plain language
- Generate severity-based risk scores or classifications
- Improve transparency and informed decision-making for users

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Suggested Features

Participants may choose to implement features such as:

- Clause extraction and classification
- Contract risk scoring systems
- Adversarial legal reasoning workflows
- Liability and obligation analysis
- Ambiguity and contradiction detection
- Contract comparison against standard benchmarks
- Privacy and compliance analysis
- Multi-agent reasoning systems
- Scenario-based consequence simulation
- Explainable AI-based legal insights
- Negotiation recommendation systems

Example Use Cases

Examples of possible use cases include:

- Detecting restrictive non-compete clauses in employment agreements
- Identifying hidden cancellation penalties in subscription contracts
- Highlighting broad intellectual property ownership transfers in freelance agreements
- Detecting excessive personal data collection in privacy policies
- Identifying one-sided arbitration mechanisms in platform terms and conditions
- Detecting ambiguous liability limitations in vendor agreements

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Expected System Capabilities

A strong solution may demonstrate the ability to:

- Process contracts in multiple formats such as PDF, DOCX, or scanned documents
- Perform intelligent clause extraction and semantic analysis
- Reason about contractual risks beyond keyword detection
- Explain legal implications in a user-friendly manner
- Compare clauses against common legal or industry standards
- Generate actionable and interpretable risk reports
- Support multiple categories of legal documents

Evaluation Criteria

Submissions may be evaluated based on:

Evaluation Area	Description
Legal Reasoning Quality	Ability to understand and reason about contractual implications
Risk Identification Accuracy	Effectiveness in detecting harmful or high-risk clauses
Explainability	Clarity and interpretability of generated insights
Practical Applicability	Real-world usefulness and relevance
Technical Architecture	System design, modularity, and scalability
Adaptability	Ability to support diverse contract types

Innovation	Novelty in reasoning, analysis, or workflow design
User Experience	Quality, accessibility, and usability of the interface

Constraints

- The system is not expected to replace legal professionals or provide legally binding advice
- Teams may use open-source or proprietary AI models
- Use of simulated or publicly available datasets is acceptable
- Emphasis should be placed on explainability and interpretability over opaque predictions
- Real-time deployment is optional but encouraged where feasible

Deliverables

Participants are expected to provide:

- A working prototype or demonstrable system
- System architecture documentation
- Explanation of AI models, reasoning workflows, and methodologies used
- Demonstration of risk analysis capabilities
- User interface or dashboard showcasing outputs and insights
- Presentation summarizing approach, innovation, and applicability

Recommended Technology Areas

Participants may explore technologies including:

- Natural Language Processing (NLP)
- Transformer-based Legal Language Models
- Retrieval-Augmented Generation (RAG)
- Semantic Similarity & Embedding Models
- Multi-Agent AI Systems
- Explainable AI Frameworks
- OCR & Document Parsing Pipelines
- Vector Databases and Knowledge Retrieval Systems